

2025 Popcorn Hybrid Response to Nitrogen Fertilizer – Purdue Univ.**Dan Quinn, Ph.D.****Extension Corn Specialist****Purdue University****Email:** djquinn@purdue.edu**Narciso Zapata****M.S. Graduate Research Assistant****Purdue University****Email:** nzapatab@purdue.edu

Location (s): Purdue University Agronomy Center for Research and Education (ACRE), West Lafayette, IN. Pinney Purdue Agricultural Center (PPAC), Wanatah, IN. Southeast Purdue Agricultural Center (SEPAC), Butlerville, IN.

Location Descriptions:

Research trial treatment design included randomized completed block designs with 5 replications. Plot size measured 15 ft wide (6, 30-inch corn rows) x 40 ft long. Corn planting dates ranged from May 2 – June 2, 2025. Targeted population was 34,000 seeds/ac and the hybrid used were Ag Alumni AP4003, ConAgra, and Crookham. Nitrogen fertilizer was applied either as 28% UAN or 32% UAN and was applied as 40 lbs N/ac in a 2x2 starter + remaining required N based on total N rate treatment coulters-injected as a sidedress application at all locations at ~V5 growth stage. Grain yield will be harvested from the center 2 rows at each plot using a Wintersteiger plot combine equipped with a HarvestMaster weigh system. Plant samples were taken at the V8, R1, and R6 growth stages to track nutrient uptake differences between the popcorn hybrids (plant samples are still being analyzed at the lab). In addition, grain samples will be taken prior to harvest for quality, popping, and kernel specific data.

2025 Growing Season Updates and Progress:

The majority of the research trials were planted timely and successfully established. The trial in southeast Indiana was delayed until June due to persistent spring rainfall. The location at the Purdue ACRE research farm in West Lafayette, IN did experience minor lodging due to significant wind and storms in 2025. Nitrogen responses and deficiency differences between plots were present across all research trial locations in 2025 (Figure 1). In addition, lodging and “goose necking” was more pronounced in the high nitrogen rate plots (240 lbs N/acre) in comparison to the low nitrogen rate plots (0 and 120 lbs N/acre) (Figure 1).



Figure 1. Lower canopy photos highlighting nitrogen deficiency (leaf firing) differences between different N rate treatments in West Lafayette, IN. From left to right the N rates are 0 lbs N/acre, 120 lbs N/acre, and 240 lbs N/acre.

2025 Upcoming Data Collection, Processing, and Analysis

Our team is currently working on processing all of the plant tissue samples and nutrient analysis to add to the current popcorn nutrient uptake and use database that we have started in 2023. In addition, we will continue with data collection such as lodging ratings, ear samples, grain yield, and grain moisture. The ear samples will also be processed and examined for kernel quality and popping parameters to assess these aspects to different N fertilizer rates.



Figure 2. Aerial photo of the popcorn hybrid x nitrogen fertilizer rate trial in West Lafayette, IN. 2025.

Yield Response to Applied Nitrogen Fertilizer (2025)

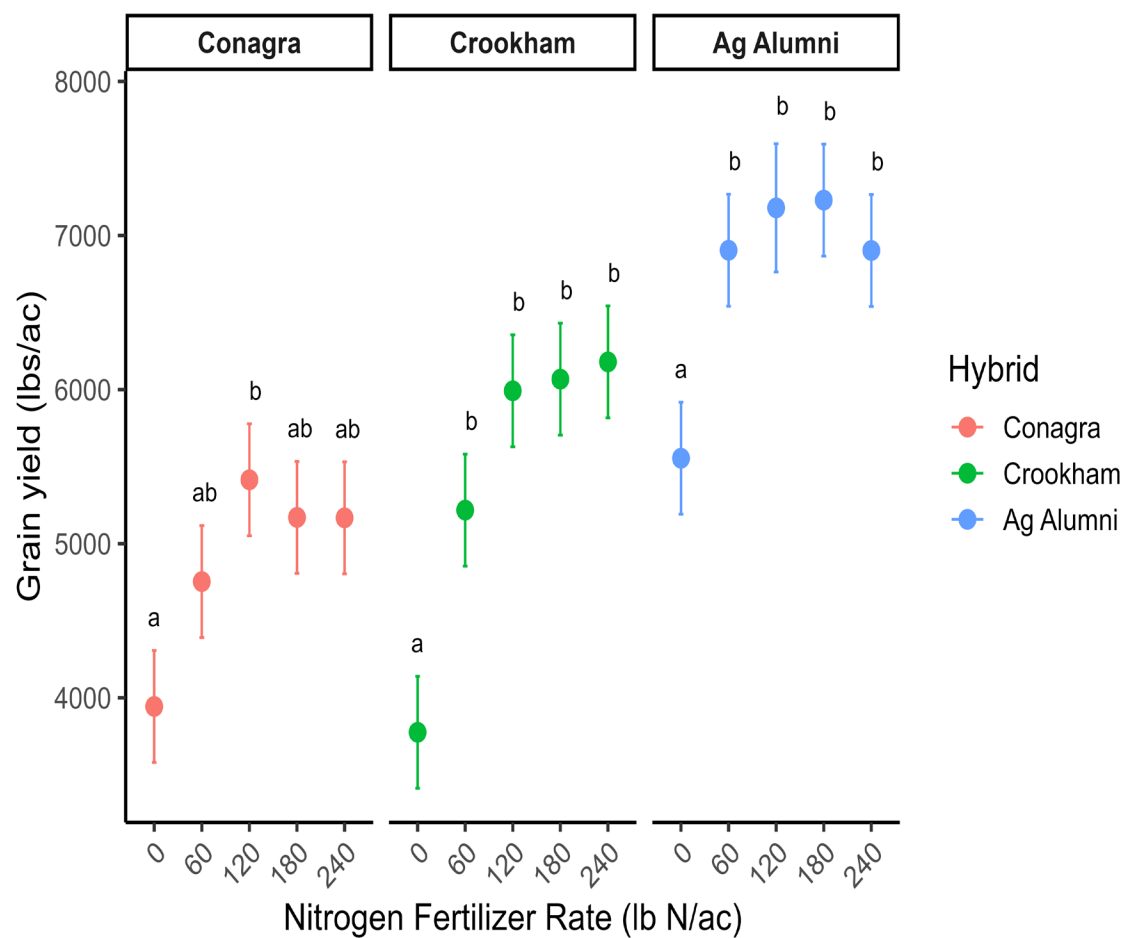


Figure 3. Popcorn grain yield response to applied total nitrogen (N) fertilizer rates. **West Lafayette, IN 2025.** Mean yield values within each hybrid that do not contain the same letter are determined statistically different ($P < 0.1$).

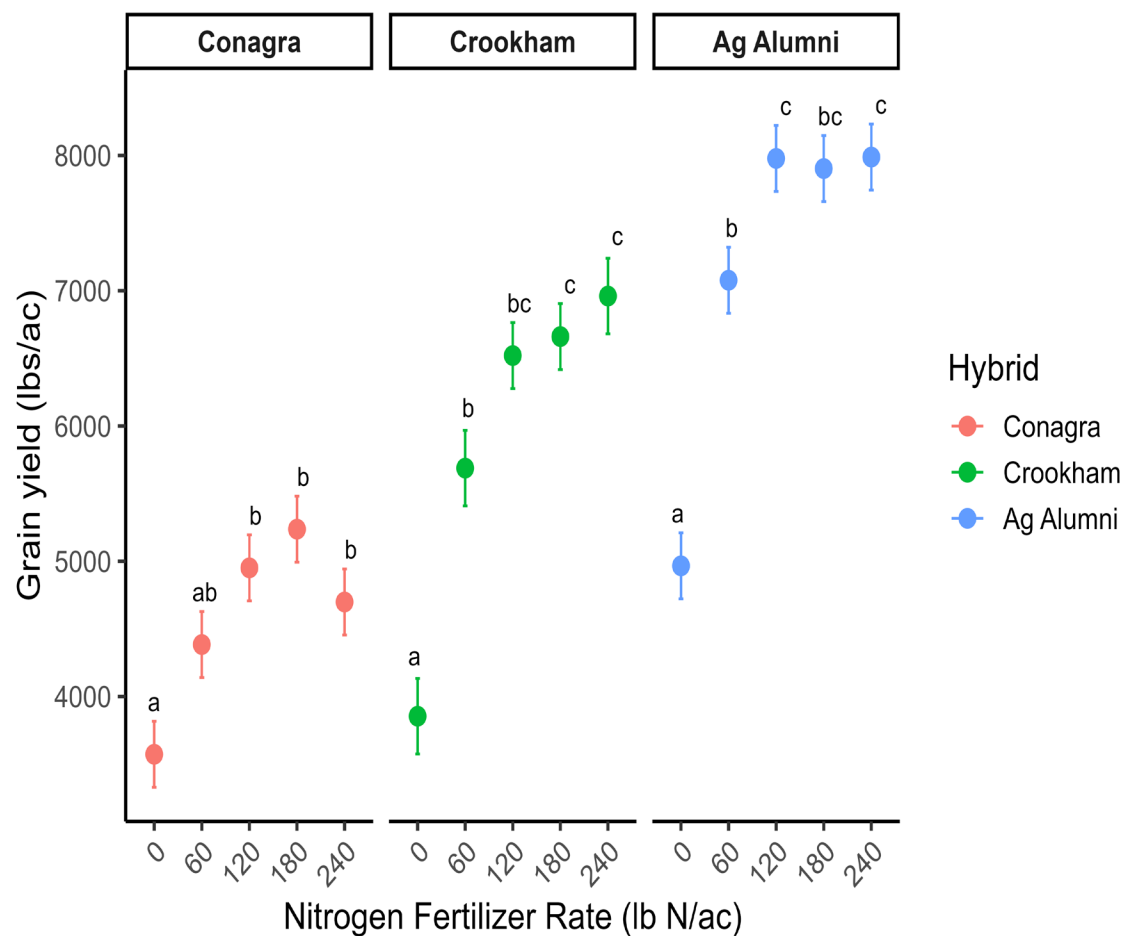


Figure 4. Popcorn grain yield response to applied total nitrogen (N) fertilizer rates. **Wanatah, IN 2025.** Mean yield values within each hybrid that do not contain the same letter are determined statistically different ($P < 0.1$).

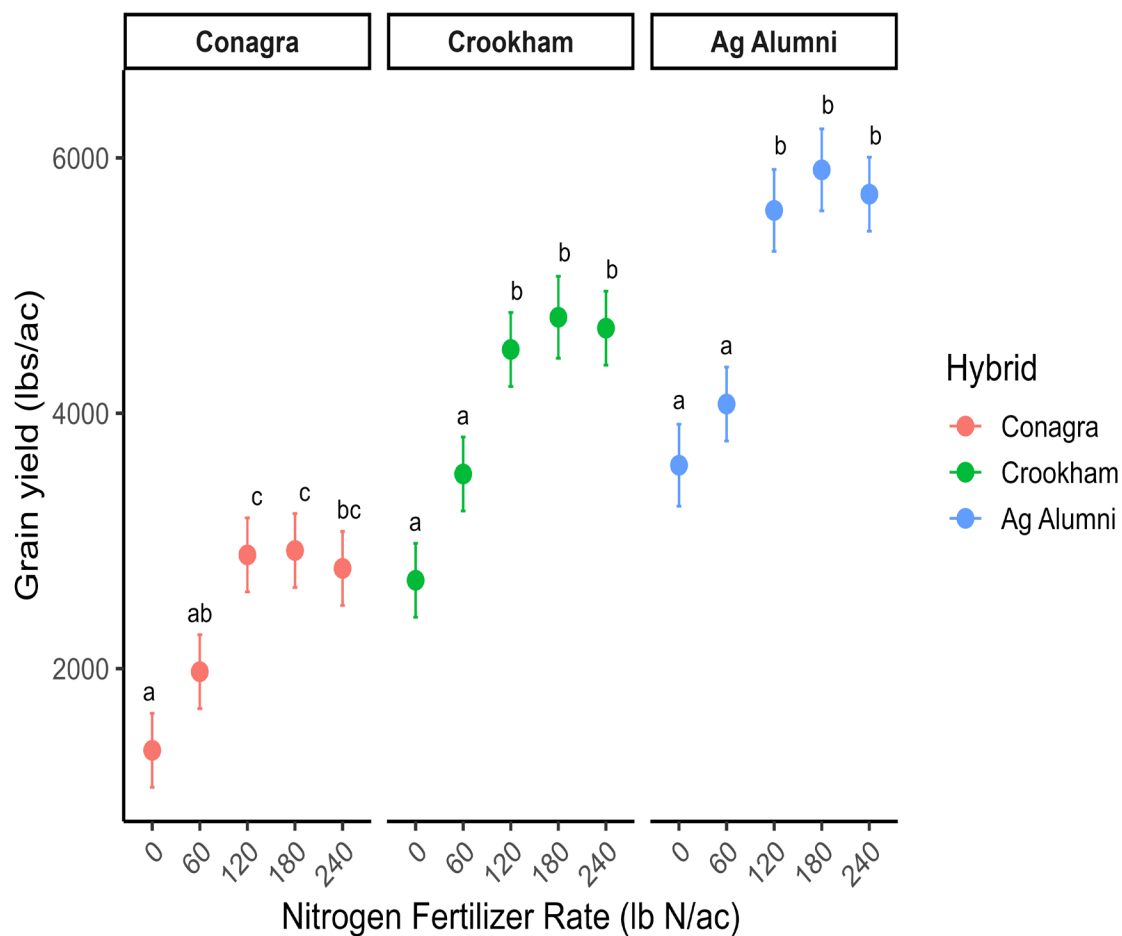


Figure 5. Popcorn grain yield response to applied total nitrogen (N) fertilizer rates. **Butlerville, IN 2025.** Mean yield values within each hybrid that do not contain the same letter are determined statistically different ($P < 0.1$).

Table 1. Quadratic plateau regression analysis parameters and calculated agronomic optimum nitrogen fertilizer rates (AONR) and yields obtained at the calculated AONR values for the different hybrids on the popcorn research trials. West Lafayette (ACRE), Wanatah (PPAC), and Butlerville (SEPAC), IN (2025).

Location	Hybrid	Model [†]	Intercept	Linear Coefficient	Quadratic Coefficient	R ²	AONR [†]	YAONR [†]
							-lbs N/ac-	--lbs/ac--
ACRE	Conagra	QP	67.82	0.34	-0.0012	0.67	135.04	5,238
	Crookham	QP	51.68	0.41	-0.0013	0.62	158.02	6,125
	Ag Alumni	QP	72.00	0.16	-0.0004	0.16	198.01	6,968
PPAC	Conagra	QP	59.79	0.30	-0.0009	0.45	159.18	4,981
	Crookham	QP	62.97	0.38	-0.0013	0.56	145.20	6,580
	Ag Alumni	QP	58.14	0.54	-0.0021	0.91	130.25	7,960
SEPAC	Conagra	QP	35.13	0.47	-0.0013	0.66	181.98	2,885
	Crookham	QP	46.79	0.37	-0.0009	0.66	204.43	4,756
	Ag Alumni	QP	47.81	0.44	-0.0011	0.69	196.28	5,839

[†] QP, quadratic-plateau regression model; YAONR, popcorn grain yield at the agronomic optimum N rate (AONR).

Post-Harvest Popcorn Kernel Data:

Table 2. Popcorn hybrid mushroom %, butterfly %, kernel size, and expansion in response to applied nitrogen (N) fertilizer rates. Data combined across all locations in the research trial. West Lafayette, Wanatah, and Butlerville, IN (2025).

Hybrid	N Rate	Mushroom	Butterfly	Kernel Size	Expansion
	--- lbs N/ac ---	--- % ---	--- % ---	--- kernels/10 g ---	--- mL/g ---
Conagra	0	24.9 a*	75.1 a*	63.6 a*	48.8 a*
	60	22.8 a	77.2 a	59.1 a	49.4 a
	120	26.4 a	73.6 a	60.5 a	49.4 a
	180	20.8 a	79.2 a	61.3 a	49.9 a
	240	25.5 a	74.5 a	61.4 a	49.5 a
Crookham	0	87.8 a	12.2 a	51.5 a	32.6 a
	60	88.0 a	12.0 a	48.5 ab	35.0 a
	120	89.7 a	10.3 a	45.7 b	34.9 a
	180	89.6 a	10.4 a	45.7 b	34.6 a
	240	89.4 a	10.6 a	46.1 ab	35.0 a
Ag Alumni	0	24.3 a	75.7 a	64.1 a	41.1 b
	60	25.2 a	74.8 a	61.9 a	43.7 ab
	120	25.6 a	74.4 a	62.8 a	44.0 ab
	180	24.9 a	75.1 a	59.8 a	46.2 a
	240	18.0 a	82.0 a	65.3 a	46.8 a

*Mean values within each individual parameter/column that do not contain the same letter are statistically different ($P < 0.1$).

Preliminary Conclusions 2025:

- Grain yield responses varied by hybrid, with Ag Alumni consistently producing the highest yields and Conagra the lowest across all locations. Within each hybrid, yield increased with applied N rate but no significant differences were observed among N rates, suggesting a clear N response that plateaus at higher N rates.
- Agronomic optimum nitrogen rate (AONR) for popcorn hybrids in 2025 ranged from 130 to 204 lb N/ac depending on hybrid and location (Table X). Nitrogen response varied substantially across environments, with stronger and more consistent quadratic-plateau fits observed at PPAC and SEPAC, while one hybrid (Ag Alumni at ACRE) showed no significant yield response to N (Figure 1). Overall, optimal N requirements were hybrid- and site-specific, emphasizing the importance of environment-specific nitrogen management strategies for popcorn production in Indiana.
- Nitrogen fertilizer rate did not significantly impact mushroom %, butterfly %, kernel size, or expansion across hybrids (Table 2). However, preliminary observations suggest potential subtle effects of N on kernel parameters and popping characteristics, particularly expansion in Ag Alumni and kernel size in Crookham hybrids.